

Sanjay Manandhar

Data Engineer | Python • SQL • Airflow • AWS • ETL

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PROFESSIONAL SUMMARY

Data Engineer with 4+ years of experience designing and maintaining data infrastructure, building ETL/ELT pipelines, and delivering analytical solutions. Skilled in Python, SQL, Apache Airflow, and AWS, with a proven track record across telecom, healthcare, public sector, and financial domains.

SKILLS

Python (NumPy, Pandas, Airflow) • SQL Server, Oracle, MongoDB • Apache Airflow, dbt, Docker • AWS (S3, Redshift, EC2) • ETL/ELT, Data Warehousing (Fact/Dimension modeling) • Apache Spark, Hadoop • Bash, Git • Data Visualization

PROFESSIONAL EXPERIENCE

DishHome Pvt. Ltd. — Data Engineer

August 29, 2025 – Present

BIR (SQL Server – Reporting & Data Warehouse)

- Designed and developed stored procedures for telecom reporting systems, including bulk/incremental refresh logic for POD performance, SME packages, and FIFA World Cup 2026 recharge/churn/bonus reporting
- Optimized high-volume SQL Server procedures by rewriting CTE-based logic into sequential temp-table approaches, eliminating full table scans on large financial transaction tables
- Built a multi-tier data reconciliation and validation engine (BNG-ICX mismatch detection) processing 300K+ records with cascading match logic
- Designed dimensional data warehouse schema (fact/dimension tables) to support enterprise-wide telecom customer and distribution reporting
- Diagnosed and resolved critical production issues in daily reporting jobs, improving pipeline reliability and data accuracy

Insights (Data Engineering – Airflow, MongoDB, Data Mart)

- Built and orchestrated end-to-end ETL pipelines using Apache Airflow, integrating Oracle, MongoDB, and SQL Server across the data platform
- Developed a MongoDB-to-SQL Server migration pipeline using Airflow and dbt, consolidating 13+ collections into staging and data mart layers for analytics
- Built cross-system reconciliation pipelines (vendor FTP-based) with automated data validation and portal upload workflows
- Designed incremental/watermark-based and CDC data pipelines to sync high-volume datasets (SMS, dealer hierarchy, entitlement data) in near real-time
- Set up and administered core Airflow infrastructure (Docker, CeleryExecutor, Oracle integration) from the ground up, establishing the ETL foundation for the platform

Cedar Gate Technologies — Data Engineer / Data Analyst

October 2023 – May 2024

- Queried and optimized large-scale healthcare datasets in Amazon Redshift, supporting performance improvement initiatives
- Mentored junior personnel on the U.S. healthcare data ecosystem and internal tooling practices
- Standardized data using in-house ETL tools and modules to produce analysis-ready outputs
- Utilized AWS S3 for data versioning and supported development of analytical software for client-facing insights
- Applied CMS Data Code Sets (ICD-10, CPT-4, HCPCS, NDC, Provider Taxonomy Codes) and HEDIS/HCC methodologies for quality and risk adjustment reporting

Young Innovations Pvt. Ltd. — Data Intern to Junior Data Professional

February 2022 – July 2023

- Processed survey data using Python and ODK, collaborating with field engineers to ensure clean, complete data collection (RVS)
- Automated data syncing and updates through ODK Central for nationwide reporting (RVS)
- Cleaned and standardized national census datasets using Python, creating analytical visualizations to support demographic research (Census Nepal)
- Uploaded processed census data to AWS S3 for public portal distribution (Census Nepal)

Hazesoft Pvt. Ltd. — Data Science Intern to Data Scientist

February 2021 – October 2021

- Designed and implemented ETL pipelines for NEPSE stock data to support real-time analysis
- Built machine learning models for classification tasks, including fake news detection and gaming behavior categorization
- Performed data cleaning, preprocessing, and feature engineering on financial and text datasets to improve model accuracy
- Conducted exploratory data analysis (EDA) to identify trends and patterns in stock market data
- Collaborated with cross-functional teams to deploy and monitor model performance

EDUCATION

Tribhuvan University / Nepal Commerce Campus, Minbhawan

Graduated 2021

Bachelor of Information Management (BIM), CGPA: 2.87

Prasadi Academy, Lalitpur

2014

Higher Secondary Education Board (HSEB), 72.3%

Sudesha High School, Lalitpur

2012

School Leaving Certificate, 76.63%

SELECTED PROJECTS

- BIR Portal — <https://portal.dishhome.com.np/>
- Insights — <https://insights.dishhome.com.np/>
- RVS Platform — <https://rvs.ndrrma.gov.np>
- Census Nepal — <https://censusnepal.cbs.gov.np>